

A combined reliability analysis approach with dimension reduction method and maximum entropy method

Gang Li · Kai Zhang

Received: 27 October 2009 / Revised: 23 June 2010 / Accepted: 14 July 2010 / Published online: 3 August 2010
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Abstract This paper presents a combined reliability analysis approach which is composed of Dimension Reduction Method (DRM) and Maximum Entropy Method (MEM). DRM has emerged as a new approach in this field with the advantages of its sensitivity-free nature and efficiency instead of searching for the most probable point (MPP). However, in some recent implementations, the Moment Based Quadrature Rule (MBQR) in the DRM was found to be numerically instable when solving a system of linear equations for the integration points. In this study, a normalized Moment Based Quadrature Rule (NMBQR) is proposed to solve this problem, which can reduce the condition number of the coefficient matrix of the linear equations considerably and improve the robustness and stableness. Based on the statistical moments obtained by DRM+NMBQR, the MEM is applied to construct the probability density function (PDF) of the response. A number of numerical examples are calculated and compared to the Monte Carlo simulation (MCS), the First Order Reliability Method (FORM), the Extended Generalized Lambda Distribution (EGLD) and Saddlepoint Approximation (SA). The results show the accuracy and efficiency of the proposed method, especially for the multimodal PDF problem and multiple design point problem.

Keywords Reliability analysis · Dimension Reduction Method (DRM) · Moment Based Quadrature Rule (MBQR) · Maximum Entropy Method (MEM)

1 Introduction

Structural reliability is defined as the probability of safety (or proper performance) of the structure over a given period of time. The ultimate purpose of reliability analysis is to compute a multi-dimensional integral over an irregular failure region. There are several options available. (1) Direct integration. It is available only in special cases due to the complexity of the integral. Moreover, the multi-dimensional joint probability density function (PDF) is hardly available in the realistic problems (Du and Chen 2000). (2) Numerical integration. Direct Monte Carlo simulation (Rubinstein 1981) is the most widely used simulation method and there also involves quasi-Monte Carlo simulation (Niederreiter and Spanier 2000), importance sampling (Melchers 1989), directional simulation (Bjerager 1988; Nie and Ellingwood 2000) etc. Nevertheless, all the simulation methods require considerably more extensive calculations than analytical methods. Hence, the simulation methods are always employed for verification of probability analysis when alternative methods are used. (3) Analytical methods. The first- and second- order reliability methods (FORM and SORM) are considered to be the most popular in the past several decades (Hasofer and Lind 1974; Rackwitz and Fiessler 1978) which are based on the first- and second-order Taylor series expansions of the performance function at the most probable point (MPP). In general, the FORM or SORM requires an iterative process using the sensitivities of the performance function to locate the MPP after the nonlinear transformation from non-normal variables into standard

G. Li (✉) · K. Zhang
State Key Laboratory of Structural Analysis of Industrial Equipment,
Department of Engineering Mechanics, Dalian University of
Technology, Dalian, 116023, China
e-mail: ligang@dlut.edu.cn

K. Zhang
e-mail: foggy canyon@163.com

normal variables. Recently, Liang et al. proposed a single loop method to locate a MPP (Liang et al. 2008). Although for many realistic problems, FORM and SORM give the reasonably accurate solutions, they tend to suffer from numerical difficulties in the MPP search and inaccuracies due to the nonlinearity of the performance function.

Moment methods are another route for structural reliability which requires neither iterations nor the computation of sensitivities (Zhao and Ono 2001; Seo and Kwak 2002). As to the calculation of statistical moments of the response, the univariate Dimension Reduction Method (DRM) was recently proposed (Rahman and Xu 2004), which transforms a multi-dimensional response function into multiple one-dimensional functions. The Moment Based Quadrature Rule (MBQR) is then carried out for the numerical integration in the DRM efficiently. The virtue of this approach over other analytical methods is that it does not require the calculation of MPP; moreover, it needs considerably fewer response function evaluations in comparison to other simulation-based or analytical methods. However, in the recent implementation of this method, the MBQR in the DRM was found to be numerically instable when solving a system of linear equations for the integration points (Youn et al. 2008; Ju and Lee 2007; Choi et al. 2007).

In this study, a Normalized Moment Based Quadrature Rule (NMBQR) is proposed to solve this difficulty. Considering that the maximum entropy method (MEM) is regarded as the most unbiased estimation for the response PDF, it is used to approximate the PDF of the output random variable after the statistical moments are determined using the DRM. A number of numerical examples are examined to show the effectiveness and accuracy of the proposed approach (DRM+MEM), compared to the Monte Carlo simulation (MCS), the First Order Reliability Method (FORM) DRM+Pearson, DRM+Extended Generalized Lambda Distribution (EGLD) and DRM+Saddlepoint Approximation (SA).

2 Review of univariate dimension reduction method

2.1 Introduction of univariate dimension reduction method

Recently, the univariate dimension reduction method (DRM) was proposed by Rahman and Xu (2004) in order to predict the statistical moments of the response function which is affected by the random variable such as the random loads, material properties etc. The main thought of this method is the reduction from a multi-dimensional integration into a number of one-dimensional integrations with the idea of additive decomposition; the approximation of response moments by moments of single random variables;

and a Moment Based Quadrature Rule (MBQR) for numerical integration. Here is the brief introduction of DRM (Rahman and Xu 2004).

In general, considering an N -dimensional, differentiable, real-valued performance function $y(\mathbf{x})$ that depends on random variable vector $\mathbf{x} = [x_1, \dots, x_N] \in \mathbb{R}^N$, its l th raw statistical moment, m_l can be calculated as

$$m_l = E[y^l(\mathbf{x})] = \int_{\mathbb{R}^N} y^l(\mathbf{x}) f_X(\mathbf{x}) d\mathbf{x}, \quad (1)$$

where E denotes the expectation operator, $f_X(\mathbf{x})$ means the joint probability density function of $\mathbf{x}$, l represents the order of the moment calculated, y stands for the response of the system. The additive decomposition method can be used to approximate the response function $y(\mathbf{x})$, which transforms a multi-dimensional function to multiple one-dimensional functions as

$$\hat{y}(\mathbf{x}) = \hat{y}(x_1, \dots, x_N) = \sum_{i=1}^N y(\mu_1, \dots, \mu_{i-1}, x_i, \mu_{i+1}, \dots, \mu_N) - (N-1)y(\mu_1, \dots, \mu_N), \quad (2)$$

where each term in the summation is a one-dimensional random function and $y(\mu_1, \dots, \mu_N)$ is the deterministic value of $y(\mathbf{x})$ calculated at the mean values of all the random variables $\boldsymbol{\mu} = [\mu_1, \dots, \mu_N]$. Substituting the approximate function of (2) in (1) gives

$$m_l \cong E[\hat{y}^l(\mathbf{x})] = E \left[\left\{ \sum_{j=1}^N Y(\mu_1, \dots, \mu_{j-1}, x_j, \mu_{j+1}, \dots, \mu_N) - (N-1)y(\mu_1, \dots, \mu_N) \right\}^l \right], \quad (3a)$$

where the one-dimensional integration in (3a) can be expressed as

$$\begin{aligned} E[y^m(\mu_1, \dots, \mu_{j-1}, x_j, \mu_{j+1}, \dots, \mu_N)] \\ = \int_{-\infty}^{\infty} y^m(\mu_1, \dots, \mu_{j-1}, x_j, \mu_{j+1}, \dots, \mu_N) f_{X_j}(x_j) dx_j, \end{aligned} \quad (3b)$$

where m means the order of moments for the one-dimensional random function $y(\mu_1, \dots, \mu_{j-1}, x_j, \mu_{j+1}, \dots, \mu_N)$ and $f_{X_j}(x_j)$ denotes the marginal probability density of X_j , which can be calculated from the known information.

Note that (3b) is only valid for independent random vector X , and a multivariate transformation such as that of Rosenblatt or Nataf can be used to transform the dependent random variables into a set of independent standard Gaussian

random variables before the application of dimension-reduction method.

At last, the moment based quadrature rule (MBQR) is introduced to perform the one-dimensional numerical integration in (3b). Integration points and weights can be calculated by solving a linear system equation that is made up of the raw statistical moments of the input random variables. Applying the MBQR, the linear system can be achieved as shown in (4)

$$\begin{bmatrix} \mu_{j,n-1} & -\mu_{j,n-2} & \mu_{j,n-3} & \dots & (-1)^{n-1} \mu_{j,0} \\ \mu_{j,n} & -\mu_{j,n-1} & \mu_{j,n-2} & \dots & (-1)^{n-1} \mu_{j,1} \\ \mu_{j,n+1} & -\mu_{j,n} & \mu_{j,n-1} & \dots & (-1)^{n-1} \mu_{j,2} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \mu_{j,2n-2} & -\mu_{j,2n-3} & \mu_{j,2n-4} & \dots & (-1)^{n-1} \mu_{j,n-1} \end{bmatrix} \times \begin{bmatrix} r_{j,1} \\ r_{j,2} \\ r_{j,3} \\ \vdots \\ r_{j,n} \end{bmatrix} = \begin{bmatrix} \mu_{j,n} \\ \mu_{j,n+1} \\ \mu_{j,n+2} \\ \vdots \\ \mu_{j,2n-1} \end{bmatrix}, \quad (4)$$

where $\mu_{j,i}$ stands for the i th raw moment for the j th input random variable and n is the number of integration points and weights. Then the integration points $x_{j,i}$, $i = 1, \dots, n$ can be obtained by solving the following polynomial equation whose coefficients is the solution of linear system (4)

$$x_j^n - r_{j,1}x_j^{n-1} + r_{j,2}x_j^{n-2} - \dots + (-1)^n r_{j,n} = 0. \quad (5)$$

After determining the integration points, the corresponding weights can be achieved as follows

$$w_{j,i} = \frac{\int_{-\infty}^{\infty} \prod_{k=1, k \neq i}^n (x_j - x_{j,k}) f_{X_j}(x_j) dx_j}{\prod_{k=1, k \neq i}^n (x_{j,i} - x_{j,k})} = \frac{\sum_{k=0}^{n-1} (-1)^k \mu_{j,n-k-1} q_{j,ik}}{\prod_{k=1, k \neq i}^n (x_{j,i} - x_{j,k})}, \quad (6)$$

$$q_{j,i0} = 1, q_{j,ik} = r_{j,k} - x_{j,i} q_{j,ik-1}$$

where $w_{j,i}$ represents the weight at the i th integration point for the j th random variable. Then the calculation of (3b) can be replaced by the following algebraic operation

$$E[y^m(\mu_1, \dots, \mu_{j-1}, x_j, \mu_{j+1}, \dots, \mu_N)] \cong \sum_{i=1}^n w_{j,i} y^m(\mu_1, \dots, \mu_{j-1}, x_{j,i}, \mu_{j+1}, \dots, \mu_N). \quad (7)$$

Note that the MBQR requires $(nN + 1)$ number of analyses for the moments calculation for the one-dimensional random function, in which N means the number of the input random variables and n stands for the number of the integration points and weights for one variable if the integration point is used in common at each one-dimensional integration. In case of moderate nonlinearity of the response function, one can get pretty accurate results with $n = 5$ and under. As indicated by Rahman and Xu (2004), the larger the number of random variables, the greater the benefit of this reduction technique. Thus, the number of the integration points and weights depends on the nonlinearity of the performance function, and we take six integration points for the exponential function, and four integration points for the polynomial function.

2.2 Remarks on the univariate dimension reduction method

The main advantages of the DRM are threefold: the replacement of a complicated multi-dimensional integral with a number of one-dimensional integrals, the approximation of moments of the performance function by moments of individual random variables, and the moment based quadrature rule (MBQR) for its good accuracy and efficiency compared with other integration methods (Youn et al. 2008), such as the trapezoidal integration, Simpson integration etc. The resultant moment equations entail evaluating N number of one-dimensional integrals, which is substantially simpler and more efficient than performing one N dimensional integration. The DRM does not require the calculation of any partial sensitivities of response or the inversion of random matrices, as compared with commonly used Taylor expansion/perturbation methods and Neumann expansion methods, respectively (Rahman and Xu 2004).

However, in the recent implementation of this method, the Moment Based Quadrature Rule (MBQR) in the DRM was found to be numerically instable when solving a system of linear equations for the integration points in (4) (Youn et al. 2008; Ju and Lee 2007; Choi et al. 2007). In general, the singularity of the system of linear equations can be determined by the condition number of its coefficient matrix. When the mean is very large and the coefficient of variance is very small, it will lead to the considerable difference among the element values of the coefficient matrix, which includes the higher order statistical raw moments. Thus, the coefficient matrix may become into an ill-conditioned matrix with a very large condition number, which may cause the singularity and numerical unsteadiness to the solution of (4). This will be discussed further in Section 3.2.

3 NMBQR

3.1 Introduction of the NMBQR

To reduce the condition number of the matrix for a stable linear equations system, a Normalized Moment Based Quadrature Rule (NMBQR) is proposed in this paper. The normalized statistical moment, instead of the raw statistical moment, is introduced to reduce the condition number so as to improve the robustness and stability of the linear equations system. To construct the NMBQR with n integration points $x_{j,i}$, $i = 1, \dots, n$ in the direction of the x_j coordinate, define a function as

$$P(u_j) = \prod_{i=1}^n \left(\left(\frac{x_j - \mu_j}{\sigma_j} \right) - \left(\frac{x_{j,i} - \mu_j}{\sigma_j} \right) \right) f_{X_j}(x_j) \\ = \prod_{i=1}^n (u_j - u_{j,i}) f_{X_j}(x_j), \quad (8)$$

where μ_j and σ_j are the mean value and standard deviation of x_j , respectively, and $f_{X_j}(x_j)$ denotes the probability density function of a random variable x_j . Equation (8) satisfies

$$\int_{-\infty}^{\infty} P(u_j)(u_j)^i dx_j = 0; \quad i = 0, 1, \dots, n-1. \quad (9)$$

If

$$r'_{j,k} = \sum_{i_1=1}^n \sum_{i_2=1, \neq i_1}^n \dots \sum_{i_k=1, \neq i_1, i_2, \dots, i_{k-1}}^n u_{j,i_1} u_{j,i_2} \dots u_{j,i_k}; \\ k = 1, 2, \dots, n, \quad (10)$$

then (9) yields a system of linear equations

$$\begin{bmatrix} \mu'_{j,n-1} & -\mu'_{j,n-2} & \mu'_{j,n-3} & \dots & (-1)^{n-1} \mu'_{j,0} \\ \mu'_{j,n} & -\mu'_{j,n-1} & \mu'_{j,n-2} & \dots & (-1)^{n-1} \mu'_{j,1} \\ \mu'_{j,n+1} & -\mu'_{j,n} & \mu'_{j,n-1} & \dots & (-1)^{n-1} \mu'_{j,2} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \mu'_{j,2n-2} & -\mu'_{j,2n-3} & \mu'_{j,2n-4} & \dots & (-1)^{n-1} \mu'_{j,n-1} \end{bmatrix} \\ \times \begin{bmatrix} r'_{j,1} \\ r'_{j,2} \\ r'_{j,3} \\ \vdots \\ r'_{j,n} \end{bmatrix} = \begin{bmatrix} \mu'_{j,n} \\ \mu'_{j,n+1} \\ \mu'_{j,n+2} \\ \vdots \\ \mu'_{j,2n-1} \end{bmatrix}, \quad (11)$$

where the coefficient matrix consists of the known normalized moments of the input random variable x_j given by

$$\mu'_{j,i} = \int_{-\infty}^{\infty} \left(\frac{x_j - \mu_j}{\sigma_j} \right)^i f_{X_j}(x_j) dx_j \\ = \int_{-\infty}^{\infty} u_j^i f_{X_j}(x_j) dx_j, \quad i = 1, \dots, n. \quad (12)$$

After solving $r'_{j,i}$ from (11), the points $u_{j,i}$, $i = 1, \dots, n$ can easily be obtained as the i th root of the following equation

$$u_j^n - r'_{j,1} u_j^{n-1} + r'_{j,2} u_j^{n-2} - \dots + (-1)^n r'_{j,n} = 0. \quad (13)$$

The transformation of $u_j = \frac{x_j - \mu_j}{\sigma_j}$ can be performed to achieve the integration points $x_{j,i}$, $i = 1, \dots, n$ after the points $u_{j,i}$, $i = 1, \dots, n$ are obtained.

Finally, the integration weights for each integration point can be determined by

$$w_{j,i} = \frac{\int_{-\infty}^{\infty} \prod_{k=1, k \neq i}^n (u_j - u_{j,k}) f_{X_j}(x_j) dx_j}{\prod_{k=1, k \neq i}^n (u_{j,i} - u_{j,k})} \\ = \frac{\sum_{k=0}^{n-1} (-1)^k \mu_{j,n-k-1} q_{j,ik}}{\prod_{k=1, k \neq i}^n (u_{j,i} - u_{j,k})}. \quad (14)$$

$$q_{j,i0} = 1, \quad q_{j,ik} = r_{j,k} - u_{j,i} q_{j,ik-1}$$

3.2 Remarks on the NMBQR

Generally the stability and robustness of the numerical problem of linear equations can be evaluated by the effect of the perturbation of the coefficient matrix or the vector on the solution. For the ill-conditioned linear equations with a large condition number of the coefficient matrix, a slight perturbation may result in large changes in the solution. Compared to MBQR, the singularity of (11) has been improved considerably using NMBQR, and the condition number of the system of linear equations is reduced to a reasonable level, which will result in the robustness and stableness to the solution of NMBQR. Herein, we take the linear equation of the 4th variable from Example 2 in Section 5 to explain further.

For the linear equation in MBQR with the raw statistical moments, $\mu x = b$, the coefficient matrix and the right side vector are

$$\mu = \begin{bmatrix} 8.391403679451969e+020 & -4.924555376027415e+015 \\ 2.892265760016147e+010 & -1.70000000009485e+005 \\ \mathbf{1.00000000005563e+000} \\ 1.431001769742359e+026 & -8.391403679451969e+020 \\ 4.924555376027415e+015 & -2.892265760016147e+010 \\ 1.70000000009485e+005 \\ 2.442209461962360e+031 & -1.431001769742359e+026 \\ 8.391403679451969e+020 & -4.924555376027415e+015 \\ 2.892265760016147e+010 \\ 4.171209924754941e+036 & -2.442209461962360e+031 \\ 1.431001769742359e+026 & -8.391403679451969e+020 \\ 4.924555376027415e+015 \\ \mathbf{7.129791095657019e+041} & -4.171209924754941e+036 \\ 2.442209461962360e+031 & -1.431001769742359e+026 \\ 8.391403679451969e+020 \end{bmatrix}$$

and

$$b = \begin{bmatrix} 1.431001769742359e+026 \\ 2.442209461962360e+031 \\ 4.171209924754941e+036 \\ 7.129791095657019e+041 \\ 1.219625254741017e+047 \end{bmatrix}.$$

The maximum and minimum values of the coefficient matrix μ are $7.129791095657019e+041$ and $1.00000000005563e+000$, respectively, and the condition number of the matrix is $6.688679540053485e+043$. The solution of this equation is

$$x = \begin{bmatrix} 8.580163071489900e+005 \\ 2.942218639251456e+011 \\ 5.040203172495630e+016 \\ 4.313349532541942e+021 \\ 1.475246336296499e+026 \end{bmatrix}.$$

Considering a little perturbation δ to the vector b , say 1% of the second element of b , then we have

$$b_{-p} = \begin{bmatrix} 1.431001769742359e+026 \\ 2.442209461962360e+031 + \mathbf{0.01} \\ \times \mathbf{2.442209461962360e+031} \\ 4.171209924754941e+036 \\ 7.129791095657019e+041 \\ 1.219625254741017e+047 \end{bmatrix}$$

And the solution of $\mu x_{-p} = b_{-p}$ is

$$x_{-p} = \begin{bmatrix} -7.367759569598954e+014 \\ -5.012957770291692e+020 \\ -1.278038102399324e+026 \\ -1.447005314888489e+031 \\ -6.138835577443999e+035 \end{bmatrix}$$

We can see that only 1% perturbation have resulted in a great change in the magnitude and even the opposite symbol between x and x_{-p} .

For the linear equation in NMBQR with the normalized statistical moments, $\mu' x' = b'$, the coefficient matrix and the right side vector are

$$\mu' = \begin{bmatrix} 3.000000000016736e+000 & \mathbf{1.679990410217782e-017} \\ 1.00000000005565e+000 & -2.572215850792057e-017 \\ 1.00000000005563e+000 \\ 1.676629214746857e-016 & -3.000000000016736e+000 \\ \mathbf{-1.679990410217782e-017} & -1.00000000005565e+000 \\ 2.572215850792057e-017 \\ 1.50000000008398e+001 & -1.676629214746857e-016 \\ 3.000000000016736e+000 & \mathbf{1.679990410217782e-017} \\ 1.00000000005565e+000 \\ -7.915697123843141e-016 & -1.50000000008398e+001 \\ 1.676629214746857e-016 & -3.000000000016736e+000 \\ \mathbf{-1.679990410217782e-017} \\ \mathbf{1.05000000005864e+002} & 7.915697123843141e-016 \\ 1.50000000008398e+001 & -1.676629214746857e-016 \\ 3.000000000016736e+000 \end{bmatrix}$$

and

$$b' = \begin{bmatrix} 1.676629214746857e-016 \\ 1.50000000008398e+001 \\ -7.915697123843141e-016 \\ 1.05000000005864e+002 \\ -2.665658667407051e-014 \end{bmatrix}.$$

The difference between the elements of the coefficient matrix of NMBQR is $1.05000000005864e+002$ and the condition number of the matrix is $3.444501637993506e+002$. The solution of this equation is

$$x' = \begin{bmatrix} 9.415509267535236e-017 \\ -9.9999999999556e+000 \\ -2.285776470432000e-015 \\ 1.49999999999858e+001 \\ 2.724805532517158e-015 \end{bmatrix}.$$

Similarly, a little perturbation δ of 1% of the second element of the vector b is considered as

$$b'_{-p} = \begin{bmatrix} 1.676629214746857e-016 \\ 1.50000000008398e+001 + \mathbf{0.01} \\ \times \mathbf{1.50000000008398e+001} \\ -7.915697123843141e-016 \\ 1.05000000005864e+002 \\ -2.665658667407051e-014 \end{bmatrix},$$

and the solution of the linear equation with the perturbation $\mu x'_p = b'_p$ is

$$x'_p = \begin{bmatrix} 8.297980810792399e-017 \\ -9.924999999999555e+000 \\ -2.203834520328986e-015 \\ 1.462499999999857e+001 \\ 2.665483633868355e-015 \end{bmatrix}$$

We could see that 1% perturbation results in a slight change between x'_p and x' . The solution of NMBQR is stable and robust, while the solution of MBQR is much sensitive to the input parameter due to the ill-conditioned equation with a large condition number caused by the great difference between the elements of the coefficient matrix.

4 Maximum entropy method

Moment methods (Zhao and Ono 2001) for structural reliability can be regarded as another route for achieving structural reliability, which evaluates the failure probability by applying the statistical moments of the performance function. In general, the first four statistical moments, which are defined as the mean, variance, skewness, and kurtosis of the distribution, respectively, are used in predicting the unknown fitting parameters in each distribution-fitting model. Approximating the distribution of a continuous random variable with its moments of finite order is a well

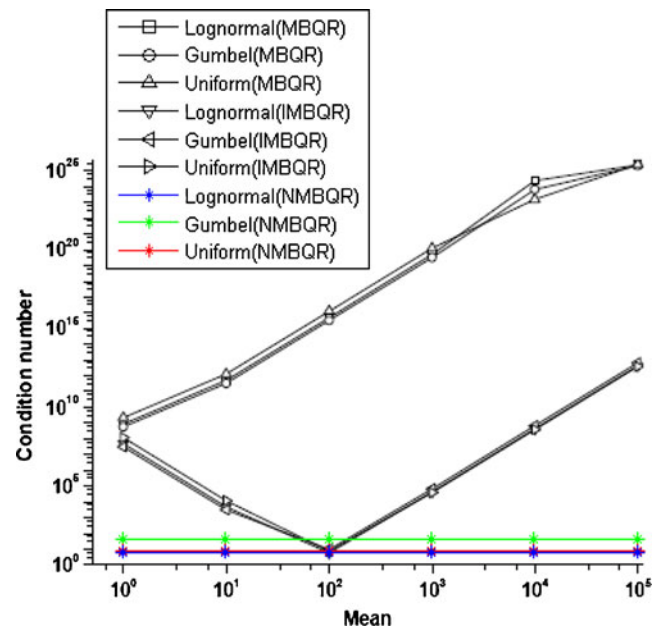


Fig. 2 Condition numbers for the variation of the mean

known problem in statistics and there are various corresponding approaches, including the Pearson and Johnson system, Saddlepoint Approximations (SA), the Extended Generalized Lambda Distribution (EGLD) and the Maximum Entropy Method (MEM). There also exist some drawbacks among these methods, for example, the Pearson

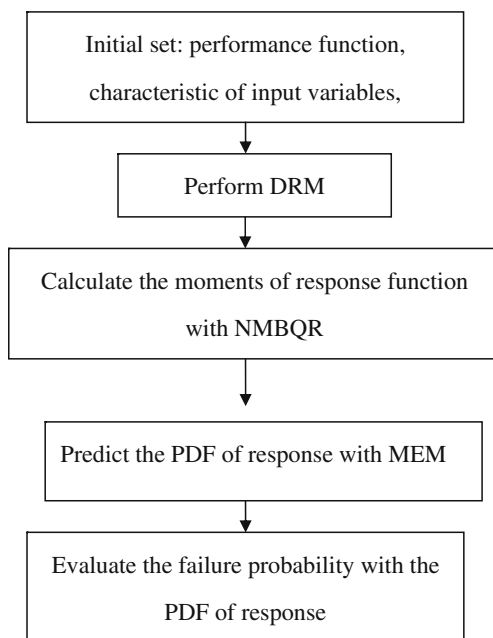


Fig. 1 Flowchart of reliability analysis with DRM+MEM

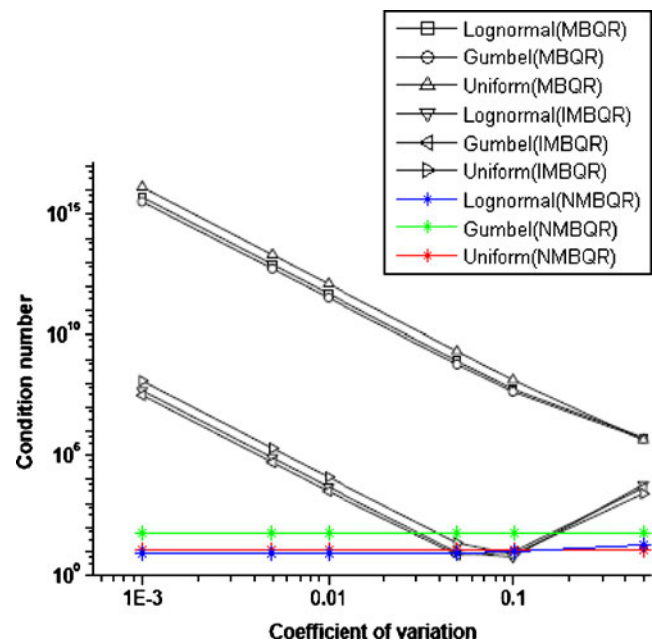
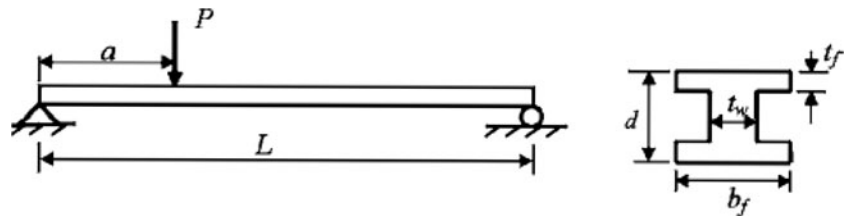


Fig. 3 Condition numbers for the variation of the coefficient of variance

Fig. 4 Simply-supported I beam

family of distributions which is widely used as the experiential function can lead to unstable results near the boundaries of different regions pertaining to different distribution types (Youn et al. 2008). It is hard to calculate the four unknown parameters from the moments of the sample data using Johnson distribution (Johnson et al. 1995). Singularities can arise during the numerical procedure in Saddlepoint approximations (Huang and Du 2006). The generalized beta and lambda distribution is just applicable for the unimodal PDF approximation of moderate to low reliability levels, and it can not predict the failure probability smaller than 0.01 (Erdem et al. 2008).

Here the maximum entropy method (MEM) is used to approximate the probability density function (PDF) of the output random variable. The MEM is regarded as the most unbiased estimation for the PDF, which means the most probable PDF from all the PDF under the moment's constraint since "it is maximally noncommittal with regard to missing information" (Papoulis and Pillai 2002) and it does not suffer the difficulties encountered with the other approaches mentioned above. As shown in the numerical examples in Section 5, the MEM could handle the multimodal PDF while other PDF approximation methods fail. Furthermore, the reliability level for the MEM approximation is still moderate low, (e.g., in Example 5, the satisfying result is obtained for the problem with failure probability of 0.002995), because very small errors in the probability distribution estimation can result in large errors in the failure probability prediction or reliability analysis (Ben Haim 1996; Ramu et al. 2007).

Table 1 Statistical properties of the random variables for Example 2

Variable	Type	Mean	Standard deviation
P	Normal	6,070	200
L	Normal	120	6
a	Normal	72	6
S	Normal	170,000	4,760
D	Normal	2.3	1/24
b_f	Normal	2.3	1/24
t_w	Normal	0.16	1/48
t_f	Normal	0.26	1/48

The MEM is a flexible and powerful tool for density approximation which maximizes the entropy $H(p) = -\sum p_i \log p_i$ under the constraints of the given moment conditions (Shannon 1948; Jaynes 1957). The MEM builds a whole family of generalized exponential distributions, including the exponential, Pareto, normal, lognormal, gamma, beta distribution as special cases which has found applications in many fields.

The MEM formulation subject to the given moment constraints for the continuous variable is defined as an optimization problem

$$\text{Maximize : } H = - \int p(x) \ln p(x) dx$$

$$\text{Subject to : } \int x^i p(x) dx = \mu_i, \quad i = 0, \dots, n, \quad (15)$$

where μ_i is the i th raw moment of the random variable, n stands for the number of the given moment constraints, always defined as 4 (Zellner and Highfield 1988; Zellner 1997); $p(x)$ denotes the PDF to be determined by the MEM.

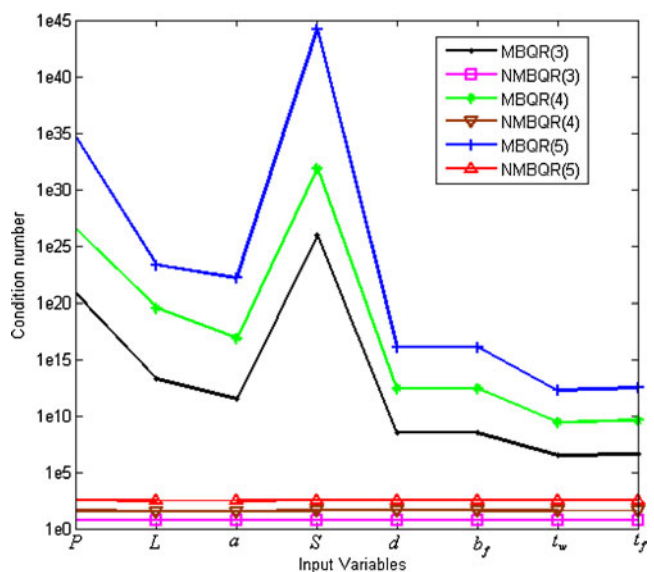
**Fig. 5** Condition numbers of the random variables

Table 2 Summary of results for Example 3

Method	No. of function evaluations	m_1	m_2	m_3	$m_4?$	P_f	Relative error (%)
DRM+MEM	$6 \times 2 + 1 = 13$	0.4317	0.3717	0.4466	0.7284	0.1127	-0.0886
DRM+SA	$6 \times 2 + 1 = 13$	0.4317	0.3717	0.4466	0.7284	0.1085	-3.81
DRM+Pearson	$6 \times 2 + 1 = 13$	0.4317	0.3717	0.4466	0.7284	0.1117	-0.98
DRM+EGLD	$6 \times 2 + 1 = 13$	0.4317	0.3717	0.4466	0.7284	0.1272	12.76
FORM	21	—	—	—	—	0.1456	29.08
MCS	1,000,000	0.4315	0.3716	0.4467	0.7334	0.1128	—

The Lagrangian multiplier λ_i , $i = 0, \dots, 4$ is used to solve this problem. The Lagrangian function can be obtained as

$$L = - \int p(x) \ln p(x) dx + \sum_{i=0}^4 \lambda_i \left(\int x^i p(x) dx - \mu_i \right). \quad (16)$$

The analytical expression of $p(x)$ is solved by $p(x) = \exp \left[- \sum_{m=0}^4 \lambda_m x^m \right]$ based on the necessary conditions for a stationary point for (16). Five Lagrangian multipliers λ_i , $i = 0, \dots, 4$ can be obtained by solving the following set of five nonlinear equations

$$G_i(\lambda) = \int x^i \exp \left[- \sum_{m=0}^4 \lambda_m x^m \right] dx = \mu_i, \quad i = 0, \dots, 4. \quad (17)$$

Here the method proposed by Zellner and Highfield (1988) is employed to calculate the Lagrangian multipliers, in which the equations are solved by the standard Newton method. First, expand the $G_i(\lambda)$ to the first order Taylor's

series around the trial λ^0 , and obtain the following linear equations

$$G_i(\lambda) \cong G_i(\lambda^0) + (\lambda - \lambda^0)^t [\text{grad} G_i(\lambda)]_{(\lambda=\lambda^0)} = \mu_i, \quad i = 0, \dots, 4. \quad (18)$$

Define the vectors δ and v as

$$\delta = \lambda - \lambda^0$$

$$v = [\mu_0 - G_0(\lambda^0), \dots, \mu_4 - G_4(\lambda^0)]^t,$$

and the Jacobian matrix G is

$$G = [g_{ij}] = \left[\frac{\partial G_i(\lambda)}{\partial \lambda_j} \right]_{(\lambda=\lambda^0)}, \quad i, j = 0, \dots, 4. \quad (19)$$

Equation (18) can be written as $G\delta = v$, which can be solved for δ when the initial value of λ^0 is given. Then update λ by $\lambda = \lambda^0 + \delta$. Note that the Jacobian matrix G is a symmetric Hankel matrix which is entirely defined by 9 values $G_i(\lambda)$, $i = 0, \dots, 8$, and we have

$$g_{ij} = g_{ji} = - \int x^i x^j \exp \left[- \sum_{m=0}^4 \lambda_m x^m \right] dx = i$$

$$= -G_{i+j}(\lambda) \quad i, j = 0, \dots, 4 \quad (20)$$

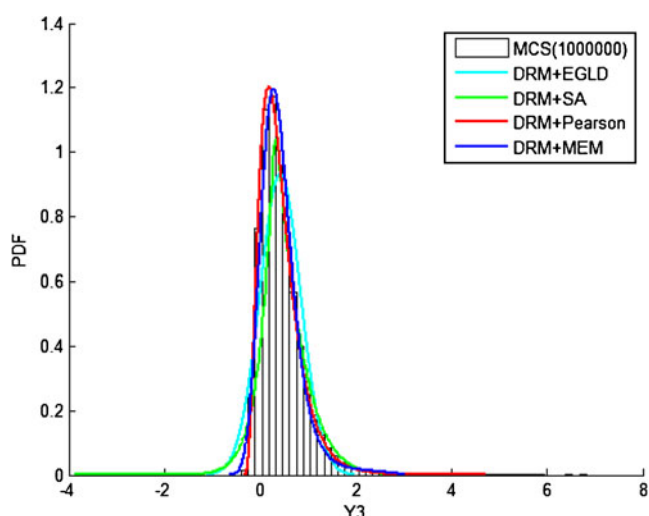


Fig. 6 Comparison of PDF of Y_3 obtained by DRM+EGLD, DRM+SA, DRM+Pearson, DRM+MEM and MCS

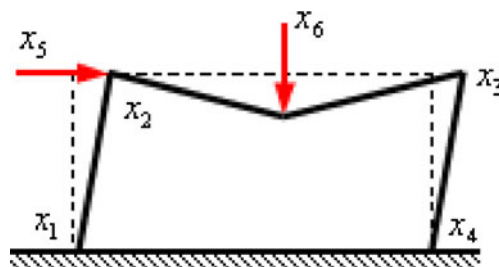


Fig. 7 A plastic collapse mechanism of one-bay frame

Table 3 Summary of results for Example 4

Method	No. of function evaluations	m_1	m_2	m_3	m_4	P_f	Relative error (%)
DRM+MEM	$4 \times 6 + 1 = 25$	270	8.3564e4	2.774e7	9.762e9	0.01243	0.566
DRM+SA	$4 \times 6 + 1 = 25$	270	8.3564e4	2.774e7	9.762e9	0.01248	0.97
DRM+Pearson	$4 \times 6 + 1 = 25$	270	8.3564e4	2.774e7	9.762e9	0.01217	-1.537
DRM+EGLD	$4 \times 6 + 1 = 25$	270	8.3564e4	2.774e7	9.762e9	0.01349	9.14
FORM	61	—	—	—	—	0.00943	-23.7
MCS	1,000,000	269.9	8.3564e4	2.7747e7	9.7682e9	0.01236	—

It is difficult to solve a large set of nonlinear integral equations in practice. So, the integrals in (17) and (20) can be treated by a univariate Simpson's method. The finite limits of the lower and upper bounds for the MEM PDF are defined as ± 10 standard deviation away from the mean. For example, we have the same result for the integral of standard normal distribution PDF on the range of $[\mu - 10\sigma, \mu + 10\sigma]$ and $[-\infty, \infty]$ considering the operational precision in Matlab language.

The procedure of the proposed combined approach of reliability analysis with DRM and MEM is showed as follows.

5 Examples

In this section, eight numerical examples taken from the published references are presented to demonstrate the effectiveness of the proposed method. Examples 1 and 2 are used to show the enhanced performance of NMBQR to

reduce the condition number considerably. Examples 3–8 compare the results of the failure probabilities obtained by the DRM+MEM, DRM+Pearson, DRM+EGLD, DRM+SA, FORM with that by direct Monte Carlo simulation (MCS), in which the NMBQR is applied in DRM.

The moments used in different PDF approximation methods are different. The raw statistical moments are used in the MEM and SA, while the central moments (the mean, variance, skewness, and kurtosis of the distribution) in the EGLD and Pearson. The transformations between raw moments and central moments are (Ormoneit and White 1999)

$$\begin{aligned}\mu'_1 &= \mu_1 \\ \mu'_2 &= \mu_2 - \mu_1^2 \\ \mu'_3 &= \frac{\mu_3 - 3\mu_2\mu_1 + 2\mu_1^3}{(\mu_2 - \mu_1^2)^{3/2}} \\ \mu'_4 &= \frac{\mu_4 - 4\mu_3\mu_1 + 6\mu_2\mu_1^2 - 3\mu_1^4}{(\mu_2 - \mu_1^2)^2}\end{aligned},$$

where $\mu_1, \mu_2, \mu_3, \mu_4$ are the first four raw moments and $\mu'_1, \mu'_2, \mu'_3, \mu'_4$ are the first four central moments.

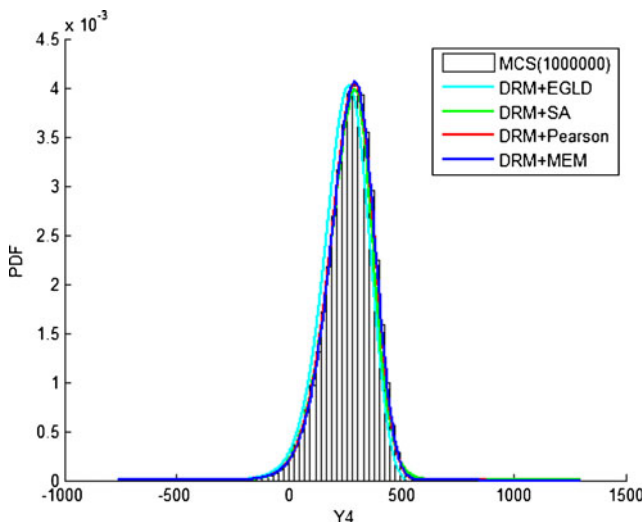


Fig. 8 Comparison of PDF of Y_4 obtained by DRM+EGLD, DRM+SA, DRM+Pearson, DRM+MEM and MCS

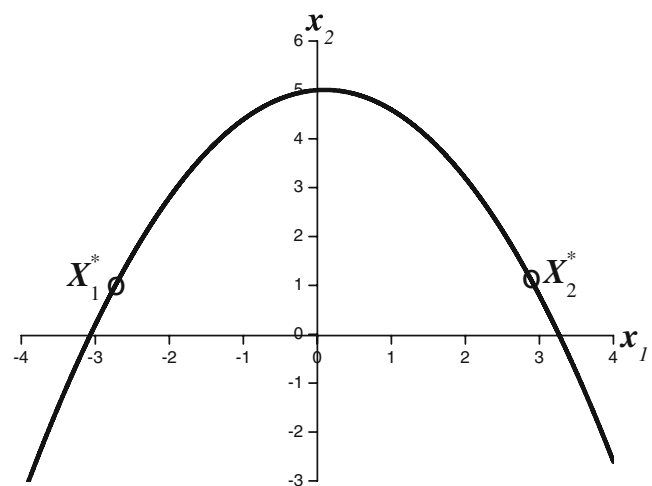


Fig. 9 The curve of the limit state function with two design points

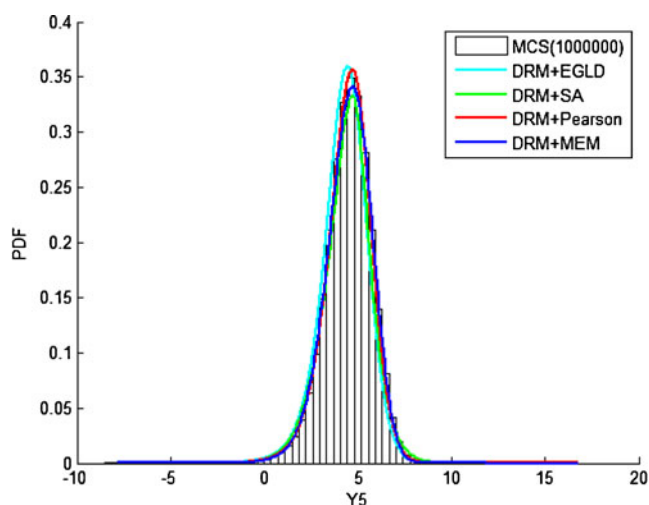
Table 4 Summary of results for Example 5

Method	No. of function evaluations	m_1	m_2	m_3	m_4	P_f	Relative error (%)
DRM+MEM	$4 \times 2 + 1 = 9$	4.495	21.715	110.1539	582.741	0.002916	2.6377
DRM+SA	$4 \times 2 + 1 = 9$	4.495	21.715	110.1539	582.741	0.002704	-9.7162
DRM+Pearson	$4 \times 2 + 1 = 9$	4.495	21.715	110.1539	582.741	0.002738	8.58
DRM+EGLD	$4 \times 2 + 1 = 9$	4.495	21.715	110.1539	582.741	0.002537	-15.292
FORM	21/21	—	—	—	—	0.00183/ 0.0009874	38.9/67.03
MCS	1,000,000	4.4966	21.729	110.254	583.458	0.002995	—

5.1 Example 1

This example (Ju and Lee 2007) is to show the difference of the condition number obtained by MBQR, Improved Moment Based Quadrature Rule (IMBQR, Ju and Lee 2007) and NMBQR, respectively. Given three distributions, lognormal, Gumbel, and uniform distribution, the condition number is calculated with three integration points and weights when the mean or the coefficient of variation is changed.

To compare the condition numbers obtained by MBQR, IMBQR and NMBQR better, Figs. 2 and 3 are adapted from the reference (Ju and Lee 2007), on which the results of NMBQR are added. In Fig. 2, the coefficient of variation is fixed as 0.01 when the mean is changed. From Fig. 2, the condition numbers of the NMBQR are much smaller and more stable than those of the MBQR and IMBQR. The condition number of the MBQR increases as the mean increases and the condition number of the IMBQR can also be unstable if the mean is changed drastically. The similar conclusion can be obtained from Fig. 3, in which the mean is fixed as 10 when the coefficient of variation is changed.

**Fig. 10** Comparison of PDF of Y_5 obtained by DRM+EGLD, DRM+SA, DRM+Pearson, DRM+MEM and MCS

Therefore, the MBQR will produce a very large condition number when the mean is very large and the coefficient of variance is very small. The IMBQR solved this problem by imposing an optimization procedure to minimize the condition number which needs more additional computational efforts, but the great variance of the conditional number may still occur in the situation mentioned in the IMBQR.

5.2 Example 2

A simply-supported I-beam (Huang and Du 2006) is calculated to demonstrate the effectiveness of the NMBQR, compared to the MBQR. The beam is acted on a concentrate force P with a distance a away from the fixed end as shown in Fig. 4.

The maximum stress due to bending can be expressed as

$$\sigma_{\max} = \frac{Pa(L-a)d}{2LI} \left(I = \frac{b_f d^3 - (b_f - t_w)(d - 2t_f)^3}{12} \right)$$

The beam is safe only if the maximum stress is less than a target value S . A system performance function is defined as $Y = g(X) = \sigma_{\max} - S$ with the safety domain $Y < 0$. The uncertainty properties of eight random variables are shown in Table 1.

There are 8 input random variables considered in Example 2. Figure 5 gives the condition number for each input variable when solving the corresponding linear equations of (11) of the one-dimensional random function, in which the x-axis in Fig. 5 represents the eight specific input variables. The number of the integration points and weights depends on the nonlinearity of the performance function, and three kinds of cases with three, four and five integration points are discussed herein, respectively. From Fig. 5, the condition numbers of MBQR are much larger than those of NMBQR because of the large mean and small coefficient of variance. With the increment of the integration points, the condition numbers of the MBQR increase greatly while those of the NMBQR grow slightly. There is a considerable difference

Table 5 Summary of results for Example 6

Method	No. of function evaluations	m_1	m_2	m_3	m_4	P_f	Relative error (%)
DRM+MEM	$4 \times 5 + 1 = 21$	1,604.9	3.199e6	7.1339e9	1.7427e13	0.02108	4.719
DRM+SA	$4 \times 5 + 1 = 21$	1,604.9	3.199e6	7.1339e9	1.7427e13	0.02110	4.818
DRM+Pearson	$4 \times 5 + 1 = 21$	1,604.9	3.199e6	7.1339e9	1.7427e13	–	–
DRM+EGLD	$4 \times 5 + 1 = 21$	1,604.9	3.199e6	7.1339e9	1.7427e13	0.02158	7.203
FORM	61	–	–	–	–	6.14e-004	–96.949
MCS	1,000,000	1,603.1	3.193e6	7.1338e9	1.7491e13	0.02013	–

of the condition number corresponding to each input variable in the MBQR while nearly the same condition number in the NMBQR.

5.3 Example 3

A concave function (Rahman and Xu 2004; Erdem et al. 2008) is used to compare the accuracy and efficiency of different reliability analysis methods, including DRM+MEM, DRM+Pearson, DRM+EGLD, DRM+SA, FORM, MCS, etc.

$$Y_3 = \frac{1}{10} [\exp(0.8X_1 - 1.2) + \exp(0.7X_2 - 0.6) - 5]$$

where X_j , $j = 1, 2$ are statistically independent and normally distributed with the mean of 3 and the standard deviation of 0.8. The MCS is conducted with 1,000,000 samples, while the DRM+MEM and DRM+Pearson, DRM+EGLD and DRM+SA operate with $(nN + 1)$ number of function evaluations, in which n is the number of points and weights and N is the number of random variables. Table 2 provides the summary of results, including the predicted values for the first four raw moments of Y_3 , m_i , $i = 1, 2, 3, 4$, the number of function evaluations and the failure probability P_f . Thirteen function evaluations are needed for the methods of DRM+MEM, DRM+EGLD, DRM+SA and DRM+Pearson, less than that of FORM with 21 function evaluations. Furthermore, the failure probabilities are very well predicted by DRM+MEM, compared with that obtained by MCS, and the results of DRM+Pearson and DRM+SA are also accepted. The failure probability estimated by FORM has an obvious error with more computation effort. The PDFs of Y_3 from different methods are shown in Fig. 6.

5.4 Example 4

A plastic collapse mechanism of one-bay frame from Zhao and Ono (2001) is considered, as shown in Fig. 7.

$$Y_4 = x_1 + 2x_2 + 2x_3 + x_4 - 5x_5 - 5x_6$$

where, the random variables x_i mean the resistance of the plastic hinge and external loads which are statistically independent and lognormally distributed with the mean and standard deviation as $\mu_1 = \dots = \mu_4 = 120$, $\mu_5 = 50$, $\mu_6 = 40$ and $\sigma_1 = \dots = \sigma_4 = \sigma_6 = 12$, $\sigma_5 = 15$. Four integral points are taken to the calculation. The results in Table 3 indicate that the DRM+MEM, DRM+EGLD, DRM+SA and DRM+Pearson are much better than that of FORM with less calculation effort in the non-normal distributions, in which DRM+MEM has the least error compared to MCS. The PDFs of Y_4 from different methods are also depicted in Fig. 8.

5.5 Example 5

This parabolic performance function example is from Zhao and Ono (2001), which was proposed by Der Kiureghian and Dakessian (1998).

$$Y_5 = b - x_2 - k(x_1 - e)^2$$

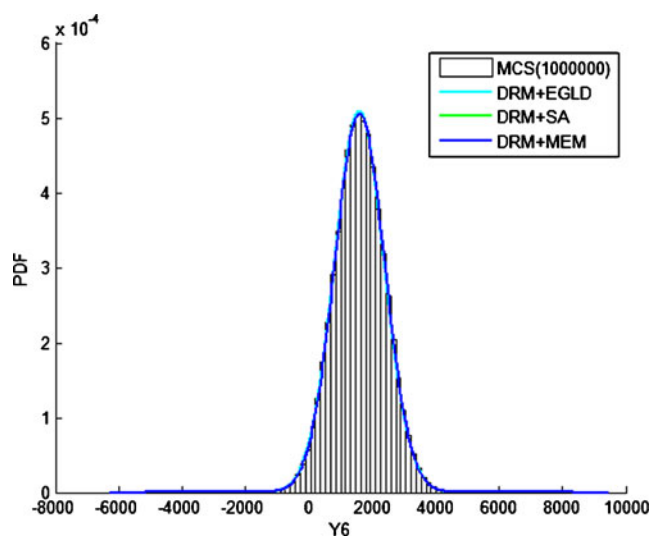


Fig. 11 Comparison of PDF of Y_6 obtained by DRM+EGLD, DRM+SA, DRM+Pearson, DRM+MEM and MCS

where x_1 and x_2 are standard normal variables, $b = 5$, $k = 0.5$ and $e = 0.1$. There exist two design points and reliability indexes as $x_1^* = [-2.741, 0.965]^T$ and $\beta_1 = 2.906$, $x_2^* = [2.916, 1.036]^T$ and $\beta_2 = 3.094$, which are shown in Fig. 9. Table 4 gives the result for Example 5. The result of DRM+MEM has a good accuracy and low burden of computation, compared to the other results of DRM+SA, DRM+Pearson, DRM+EGLD and FORM which have relatively large error. The PDF comparisons are shown in the Fig. 10.

5.6 Example 6

The performance function (Tong and Zhao 1997) is given as

$$Y_6 = 1000x_1x_2 - 7.51x_1x_3 + x_1x_4 + 40x_5 - 0.5x_3^2$$

where x_i is statistically independent and normally distributed with the means

$$\mu_1 = 1.2, \mu_2 = 2.4, \mu_3 = 50, \mu_4 = 25, \mu_5 = 10,$$

and standard deviations

$$\sigma_1 = 0.36, \sigma_2 = 0.072, \sigma_3 = 3, \sigma_4 = 7.5, \sigma_5 = 5.$$

From the results in the Table 5, the DRM+Pearson fails to predict the PDF while the DRM+MEM assesses the PDF well based on the moments from DRM which is more efficient and more accurate than that from FORM. The DRM+EGLD, DRM+SA and DRM+MEM also perform good PDF approximations. Figure 11 gives the comparison of PDF of Y_6 obtained by different methods.

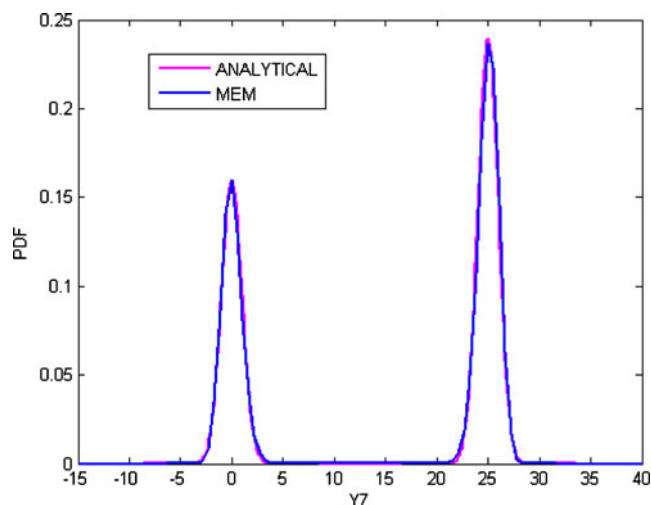


Fig. 12 Comparison of PDF of Y_7 obtained by MEM and Analytical

Table 6 Summary of results for Example 7

	MEM	Analytical	Relative error (%)
m_1	15.0000	15	0
m_2	376.0001	376	2.6596e-5
m_3	9,420.0041	9,420	4.3524e-5
m_4	236,628.1421	236,628	6.0052e-5

5.7 Example 7

A bimodal PDF example is presented to compare the ability to approximate the multimodal PDF problem, which is formulated by the weighted sum of two normal PDFs as

$$Y_7 = 0.4x_1 + 0.6x_2,$$

where the random variables x_i are statistically independent and normally distributed with the means $\mu_1 = 0$, $\mu_2 = 25$ and standard deviations $\sigma_1 = \sigma_2 = 1$. It should be noted that Y_7 is a random variable with the bimodal PDF rather than the performance function. The first four raw moments and central moments can be obtained as [15, 376, 9420, 236628] and [15, 12.288211, -0.40420, 1.19087], respectively. The EGLD, SA, Pearson and MEM are used to approximate the bimodal PDF, in which EGLD, SA and Pearson fail to get the results and MEM fits the analytical results well, shown in Fig. 12. The comparisons of the raw moments calculated with PDFs obtained by MEM and the analytical expression directly, listed in the Table 6, further demonstrate the accuracy of the MEM approximation.

5.8 Example 8

The engineering example of the vehicle side impact crash problem was firstly studied by Gu et al. (2001) and then

Table 7 Statistical properties of the random variables for Example 8

Random variable	Type	Standard deviation	Mean
X_1 (B-pillar inner)	Normal	0.030	1.000
X_2 (B-pillar reinforce)	Normal	0.030	1.000
X_3 (Floor side inner)	Normal	0.030	1.000
X_4 (Cross member)	Normal	0.030	1.000
X_5 (Door beam)	Normal	0.030	1.000
X_6 (Door belt line)	Normal	0.030	1.000
X_7 (Roof rail)	Normal	0.030	1.000
X_8 (Mat. floor side inner)	Normal	0.006	0.300
X_9 (Mat. floor side)	Normal	0.006	0.300
X_{10} (Barrier height)	Normal	1.000	10.000
X_{11} (Barrier hitting)	Normal	1.000	10.000

Table 8 Summary of results for Example 8

Method	No. of function evaluations	m_1	m_2	m_3	m_4	P_f	Relative error (%)
DRM+MEM	$4 \times 7 + 1 = 29$	0.5503	0.3874	0.3062	0.2668	0.02934	-0.034
DRM+SA	$4 \times 7 + 1 = 29$	0.5503	0.3874	0.3062	0.2668	0.02936	0.034
DRM+Pearson	$4 \times 7 + 1 = 29$	0.5503	0.3874	0.3062	0.2668	0.02934	-0.034
DRM+EGLD	$4 \times 7 + 1 = 29$	0.5503	0.3874	0.3062	0.2668	0.02973	1.29
FORM	71	—	—	—	—	0.0296	0.851
MCS	1,000,000	0.5503	0.3875	0.3064	0.2672	0.02935	—

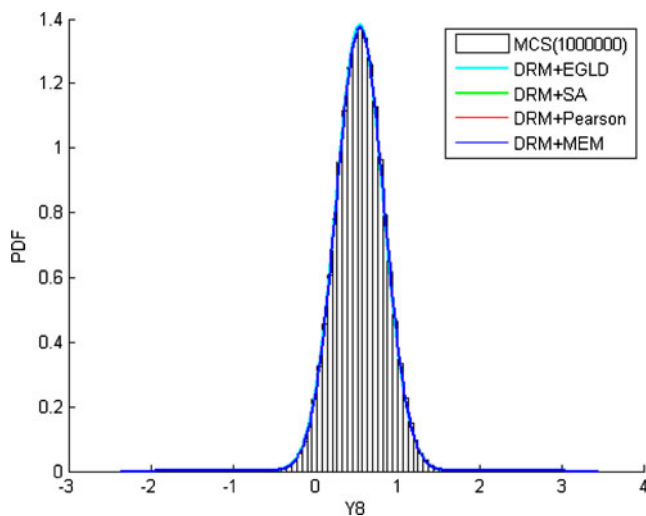
calculated by many researchers (Yang and Gu 2004; Youn et al. 2004, 2008; Yi et al. 2008). The statistic properties of the input random variables are given in Table 7. The performance of the velocity of B-pillar is studied to show the reliability estimated by the methods mentioned above, which is expressed as

$$V_{B-pillar} = 10.58 - 0.674x_1x_2 - 1.95x_2x_8 + 0.02054x_3x_{10} - 0.0198x_4x_{10} + 0.028x_6x_{10}.$$

The concerned performance function can be written as (Yi et al. 2008)

$$Y_8 = 9.9 - V_{B-pillar}$$

The results of example 8 are shown in Table 8 and their PDFs are depicted in Fig. 13 where the reliability estimations of PDF's approximation methods all have good accuracy compared to the MCS.

**Fig. 13** Comparison of PDF of Y_8 obtained by DRM+EGLD, DRM+SA, DRM+Pearson, DRM+MEM and MCS

6 Conclusions

This paper presents a combined approach for structural reliability analysis based on the univariate Dimension Reduction Method (DRM) and the Maximum Entropy Method (MEM). The Normalized Moment Based Quadrature Rule (NMBQR) is proposed to solve the singularity and numerical instability of the MBQR. The NMBQR can result in a smaller condition number for arbitrary values of mean or coefficient of variance than the MBQR and IMBQR, without any additional computational effort.

The efficiency and accuracy of the proposed DRM+MEM are studied, compared with DRM+SA, DRM+EGLD, DRM+Pearson, FORM and Monte Carlo simulation. Some observations can be obtained from the results of the numerical examples. (1) The combination of DRM+MEM, DRM+Pearson, DRM+SA and DRM+EGLD can give good results for reliability analysis in many cases. It is observed from the limited case studies in this paper that the DRM+MEM produce the best result. (2) The MEM can handle the multimodal PDF problem with satisfying results, while Pearson, SA and EGLD fail. Furthermore, the DRM+MEM can obtain much better result than other methods when treating the multiple design point problem. (3) The FORM needs more computational efforts and produces less accurate results than other PDF approximation methods in some cases, such as the high nonlinear problems and the multiple design point problems.

Therefore, the DRM+MEM is a powerful tool for PDF approximation with fewer function evaluations and sensitivity-free property for the problems of the moderate low reliability level.

Acknowledgments The supports of the National Natural Science Foundation of China (No.90815023 and No.10721062), the National Basic Research Program of China (973 Program, 2006CB705403) are much appreciated.

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